

Compact pendant with real-time fluid simulation on an LED matrix

Marharyta Paduchak, Daryna Shevchuk, Olena Dovbenchuk
Department of Computer Science
Ukrainian Catholic University
Lviv, Ukraine

Mentor: Oleksiy Hoyev
alexg.lviv@gmail.com

I. INTRODUCTION

The goal of our project is to design and build a compact wearable device — a pendant — capable of simulating real-time fluid behavior on a two-dimensional LED matrix. The system aims to visually represent how liquid would move and react to motion, tilt, or rotation in space, driven by real-time sensor data.

To achieve this, the device will be powered by a micro-controller unit (MCU) responsible for performing real-time calculations of a fluid simulation algorithm. An Accelerometer will provide data about the orientation and movement of the pendant, allowing the simulated fluid to respond naturally to the user's gestures and gravity. The visual output will be displayed on a custom-built LED matrix. The pendant will operate on battery power and be enclosed in a 3D-printed case to ensure compactness, portability, and aesthetic appeal.

II. KEY STAGES

- **Algorithm:**
 - Make simulation of the fluid handling algorithm.
 - Make implementation of the fluid handling algorithm.
 - Adapt the algorithm to work with the data from the accelerometer.
- **Hardware development:**
 - Design the LED matrix.
 - Build and implement the LED matrix.
- **Printed Circuit Board (PCB):**
 - Creating the circuit layout.
 - Manufacturing the printed circuit board.
- **Pendant's enclosure:**
 - Design 3D model of a pendant.
 - Print the pendant's enclosure.
- **Integration and testing:**
 - Combining all components into a working prototype.

III. ALGORITHM

The visual behavior of the fluid inside the pendant is modeled using a simplified numerical solution of the incompressible Navier–Stokes equations. These equations describe the conservation of momentum and mass in a fluid:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{1}{\rho} \nabla p + \mathbf{g}, \quad \nabla \cdot \mathbf{v} = 0$$

where $\mathbf{v}$ is the velocity field, p is the pressure, ρ is density, and $\mathbf{g}$ is gravity. In our simulation we assume the fluid is *incompressible* ($\nabla \cdot \mathbf{v} = 0$) and *inviscid* ($\mu = 0$), which simplifies the model while keeping it visually plausible in real time.

A. FLIP method

The simulation uses the FLIP (fluid-implicit particle) algorithm, a hybrid of Eulerian and Lagrangian representations:

- The Lagrangian layer consists of particles carrying velocity and position data, allowing the preservation of small-scale flow details such as splashes or swirls.
- The Eulerian grid enforces physical laws like incompressibility and pressure balance, ensuring stability.

Each time step includes five main stages:

- 1) **Particle advection:** particles move according to gravity and their velocity.
- 2) **Velocity transfer (particles → grid):** particle velocities are interpolated to the grid nodes using bilinear weighting.
- 3) **Pressure projection:** the Gauss–Seidel iterative method solves the Poisson equation for pressure to enforce $\nabla \cdot \mathbf{v} = 0$.
- 4) **Velocity correction (grid → particles):** each particle's velocity is updated by adding only the velocity change caused by pressure forces, preserving turbulence and fine details.
- 5) **Drift compensation:** the particle density field is monitored and corrected to prevent clumping or spreading due to numerical errors.

B. Numerical integration

The motion of each particle follows Newton's second law:

$$m \frac{d\mathbf{v}}{dt} = \mathbf{F}_{gravity} + \mathbf{F}_{pressure} + \mathbf{F}_{collision}.$$

C. Collision handling

The pendant's interior boundary is modeled as a circular wall. When a particle approaches the wall closer than its radius

r , its position is corrected to lie exactly on the boundary, and its velocity component along the surface normal $\mathbf{n}$ is reflected:

$$\mathbf{v}' = \mathbf{v} - 1.5(\mathbf{v} \cdot \mathbf{n})\mathbf{n}.$$

The coefficient 1.5 represents a partially elastic rebound, introducing slight energy loss to prevent perpetual oscillations and simulate realistic damping.

D. Grid-particle velocity exchange

Because particles do not form a regular structure, differential operators such as divergence $\nabla \cdot \mathbf{v}$ or gradient ∇p cannot be computed directly. Hence, we transfer particle velocities onto a uniform grid (Eulerian domain). Each grid node accumulates nearby particle velocities weighted by distance, producing a continuous velocity field:

$$\mathbf{v}_{grid}(i, j) = \frac{\sum_k w_{k,i,j} \mathbf{v}_k}{\sum_k w_{k,i,j}},$$

where $w_{k,i,j}$ is a bilinear weight inversely proportional to the particle-cell distance.

E. Pressure projection

To enforce incompressibility, we solve the Poisson pressure equation:

$$\nabla^2 p = \frac{\rho}{\Delta t} \nabla \cdot \mathbf{v},$$

using the Gauss-Seidel iterative relaxation method with boundary conditions $s_{i,j} = 0$ for solid cells. This step redistributes pressure so that any excess inflow or outflow in the velocity field is canceled.

F. Velocity to particles

After the pressure projection, each particle's velocity is corrected using the grid-based velocity change:

$$\mathbf{v}_{new} = \mathbf{v}_{old} + \alpha(\mathbf{v}_{grid}^{new} - \mathbf{v}_{grid}^{old}),$$

where α is a blending coefficient (typically 0.9 for FLIP, 0.1 for PIC) that smooths noise while preserving dynamic detail. This hybridization ensures both stability and visually rich motion.

G. Drift correction and density maintenance

Over time, accumulated numerical errors can cause particle clusters or voids, leading to non-uniform density. To mitigate this, the simulation periodically measures the number of particles per grid cell and applies a corrective "pressure" force whenever the local density deviates from its initial value:

$$\Delta \mathbf{v} \propto (\rho_{cell} - \rho_0)\mathbf{n}.$$

This restores the equilibrium and maintains a cohesive, volume-preserving flow even after long simulation runs.

H. Visualization mapping

Each particle influences a subset of LEDs within its local radius. The brightness of each LED is determined by the interpolated density of nearby particles, producing a smooth luminous pattern resembling liquid motion. Thus, the LED ring acts as a visual projection of the fluid density field, directly mapping physics to light intensity.

IV. PENDANT-CASE (3D MODEL)

The enclosure of the pendant was designed as a compact three-part 3D model and prepared for additive manufacturing. The geometry was optimized to: (i) fit two circular PCBs, (ii) provide a clear optical window above the LED matrix, (iii) protect the electronics from mechanical stress, and (iv) allow convenient charging access.

A. Mechanical structure

The case consists of three printed elements:

- **Top lid (dome holder):** front element with a recess for the transparent dome/glass, defining the optical aperture and keeping the LED matrix aligned with the center.
- **Brim (main body/spacer):** middle ring that sets the internal height, holds the PCBs, and forms the outer wall of the pendant.
- **Bottom lid:** rear cover that closes the enclosure and provides access openings (charging port) and a mechanical support for the battery area.

A small loop is integrated into the body for attaching a chain. All edges were rounded to improve wearing comfort and reduce stress concentration after printing.

B. Assembly considerations

The LED matrix PCB is placed directly under the transparent dome to minimize light loss and maximize contrast. The main electronics PCB is positioned behind it, separated by the spacer ring. The case provides mechanical stops that prevent board rotation and maintain alignment between the LED matrix and the enclosure center. The charging connector is aligned with the port cutout to allow charging without disassembly.

C. Manufacturing

All parts were exported as STL files and prepared for 3D printing. The model avoids steep unsupported overhangs in the critical optical area, reducing post-processing needs. After printing, the transparent dome is inserted into the recess of the top lid, and the enclosure is closed by the bottom lid.

V. CHARLIEPLEXING ALGORITHM IN LED MATRIX CONTROL

A. Concept Overview

Charlieplexing is a multiplexing technique designed to control a large number of light-emitting diodes (LEDs) using a minimal number of microcontroller input/output (I/O) pins. The method takes advantage of two properties of LEDs:

- 1) Their ability to conduct current only in one direction (from anode to cathode).
- 2) The capability of microcontroller pins to operate in three distinct states: logic HIGH, logic LOW, and high-impedance (Hi-Z).

Using these principles, the Charlieplexing technique enables each pair of microcontroller pins to address one LED in each direction. If the system uses N pins, the total number of controllable LEDs is:

$$N \times (N - 1)$$

This relationship demonstrates that the number of addressable LEDs grows quadratically with the number of available pins, allowing for efficient scaling.

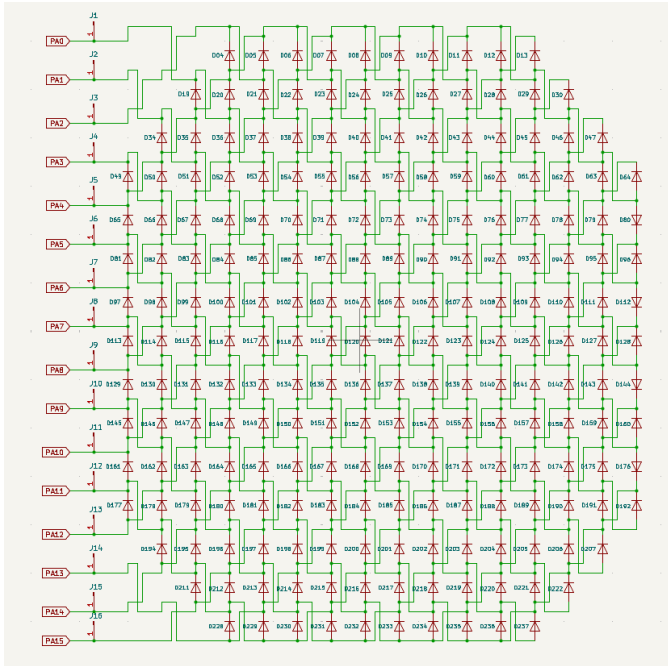


Fig. 1. Charlieplexing.

B. Operating Principle

Each LED in a Charlieplexed network is connected between two unique I/O pins, with the correct polarity defining its conducting direction. To illuminate a specific LED, one pin is configured as an output HIGH (anode), another as an output LOW (cathode), and all remaining pins are set to the high-impedance state. In this configuration, current flows only through the targeted LED, while all others remain inactive.

To create the illusion of simultaneous illumination, the microcontroller rapidly cycles through the LEDs, activating one at a time with a switching frequency 100 Hz. Due to the persistence of human vision, the observer perceives all LEDs as continuously lit even though only one LED is energized at any given moment.

C. Advantages Over Conventional Multiplexing

In conventional row-column LED matrices, the number of required control lines equals the sum of rows and columns ($m + n$ for an $m \times n$ matrix).

Charlieplexing, in contrast, requires only N pins to drive up to $N \times (N - 1)$ LEDs, substantially reducing hardware complexity.

The main advantages of the Charlieplexing approach include:

- **Reduced pin count:** A small number of I/O pins can control many LEDs.
- **Lower component count:** No need for dedicated row/column drivers or transistor arrays.
- **Compact circuit layout:** Suitable for miniaturized and wearable electronic systems.

D. PCB Design and Layout

The electronic part of the pendant is realized as a pair of circular two-layer FR-4 PCBs with a diameter slightly smaller than the inner diameter of the 3D-printed case (about 28 mm). The front board carries only the dense LED matrix, while the rear board hosts the microcontroller, accelerometer, power-management circuitry and the USB connector. Boards are connected by short wires or pin headers placed near the perimeter so that they do not block the light from the LEDs.

The main design constraints for the layout were:

- fitting all components into the circular outline of the pendant;
- achieving short and balanced routes for the LED driving lines;
- providing low-noise power distribution for the MCU, sensor and analog parts;
- maintaining manufacturability with standard PCB design rules (track width and clearance on the order of 6 mil).

E. Main electronics PCB

The main PCB is organized around the low-power STM32 microcontroller placed near the center of the circle. This central placement minimizes the average length of signal traces and naturally leads to a “star”-like routing topology.

1) Component placement:

- The microcontroller U1 is in the geometrical center. All its decoupling capacitors (C2–C4, C8, C9) are located as close as possible to the corresponding VDD pins and connected with short, wide traces to a solid ground plane.
- The accelerometer U3 is placed on one side of the MCU so that its axes are clearly aligned with the pendant geometry, simplifying the interpretation of acceleration data in firmware.
- The USB connector and the Li-ion charging circuitry (U2, U4, D1, associated passives) are grouped near the bottom of the board. This keeps the high-current charging path short and provides a convenient location for the physical charging port in the enclosure.
- Single-pin connectors J1–J16 for the LED matrix are distributed around the circumference of the board. Their

angular positions match the connectors on the LED board, which simplifies wiring and avoids sharp cable bends.

- Debug and programming connectors (SWD, power test pads) are kept near the edge for easy probe access during development.

2) *Power distribution*: The pendant is powered by a single-cell Li-ion battery (VBAT). A linear charger MCP73832 (U2) charges the battery from the 5 V USB input. A low-dropout regulator (U8) then generates a 3.3 V rail used by the microcontroller, accelerometer and other digital logic.

On the PCB, the power system is implemented with the following layout decisions:

- The 3.3 V net is routed as a short, wide trace from the regulator output, branching towards the MCU and peripherals. Local bypass capacitors are placed very close to each power pin.
- Both layers of the board contain ground pours stitched together with multiple vias. Under the MCU and the accelerometer, the copper is kept as solid as possible to provide a low-impedance reference plane and to shield sensitive analog nodes.
- The USB D+ and D lines are routed as a short, tightly coupled differential pair from the Micro-USB connector to the ESD-protection device U4 and then to the MCU pins. Their length and symmetry are sufficient for the low-speed USB mode used in this design.

3) *Charlieplexing lines routing*: Each LED driving line (D1–D16) leaves the MCU, passes through its dedicated current-limiting resistor and continues to a corresponding single-pin connector on the edge of the PCB. These traces are made slightly wider than simple logic signals to reduce voltage drop when driving LEDs.

To keep the layout clean:

- routes to opposite sides of the board are fanned out on the top layer, while longer connections cross underneath on the bottom layer;
- vias are used symmetrically so that the electrical and visual characteristics of different lines are as similar as possible;
- the D1–D16 labels on the silkscreen match the logical numbering in the firmware, simplifying debugging and assembly.

F. LED matrix PCB

The LED matrix PCB is another circular two-layer board that sits directly under the transparent dome of the pendant. Its purpose is purely optical and electrical: to arrange a high number of LEDs in a visually uniform pattern and connect them according to the Charlieplexing scheme.

1) *LED arrangement*: Surface-mount LEDs are placed in a dense quasi-rectangular grid that fits inside the circular outline. The spacing between neighboring LEDs is kept uniform in both directions, creating a visually continuous “pixel” field for the fluid simulation.

Although the electrical connections follow the Charlieplexing graph, the physical placement of LEDs is optimized for

aesthetics rather than for wiring simplicity. As a result, many traces run diagonally or snake between components, especially near the center of the board.

2) *Routing strategy*: To keep the routing manageable on only two layers:

- Each LED is connected between exactly two nodes D_i and D_j . All traces related to a particular node tend to start from the corresponding connector at the board edge and then branch off into the interior.
- Where possible, traces assigned to one node occupy a “band” or “sector” of the board, minimizing crossings with other nets.
- The majority of signal routing is performed on the top layer alongside the LEDs, while the bottom layer is used for longer “express” connections and for stitching ground islands.
- Track widths and clearances respect standard manufacturing rules, but are pushed towards the lower limit to fit the large number of connections inside the small circular area.

Since only one LED is on at a time, the instantaneous current in any given trace remains low. Therefore, the narrow tracks used in the matrix are sufficient from a thermal and reliability perspective.

G. Grounding, noise and reliability considerations

Although the project is primarily visual, the correctness of the fluid simulation depends on stable readings from the accelerometer and on glitch-free operation of the microcontroller. To support this:

- Analog reference and accelerometer supply pins are decoupled separately and routed away from fast-switching LED lines.
- The ground connection between the two boards is implemented with multiple wires/pads spaced along the edge, reducing voltage differences caused by the LED scanning currents.
- All external connectors include clearly marked polarity and pinout on the silkscreen to avoid assembly mistakes.

The final layouts of both PCBs are shown in Fig. 2 and Fig. 3.

H. Schematic overview

The complete schematic of the system is summarized in Fig. 4. It contains the following functional blocks:

- the STM32 microcontroller with decoupling network and SWD programming header;
- the LIS2DE12 accelerometer with I²C pull-up resistors and interrupt line;
- the Li-ion charging circuit and battery connector;
- the 3.3 V voltage regulator;
- the USB interface with ESD protection;
- the LED driver network: resistors R5–R20 and connectors J1–J16 providing Charlieplexing lines D1–D16.

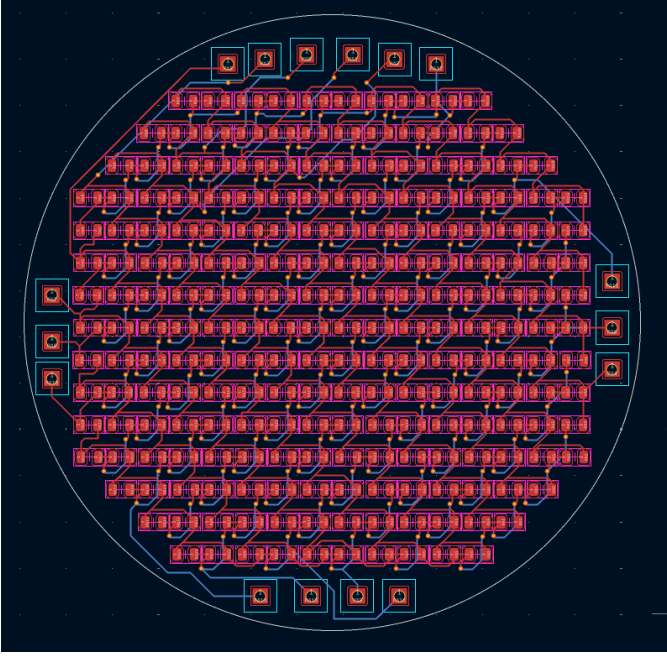


Fig. 2. Top view of the circular LED matrix PCB implementing the Charlieplexed display.

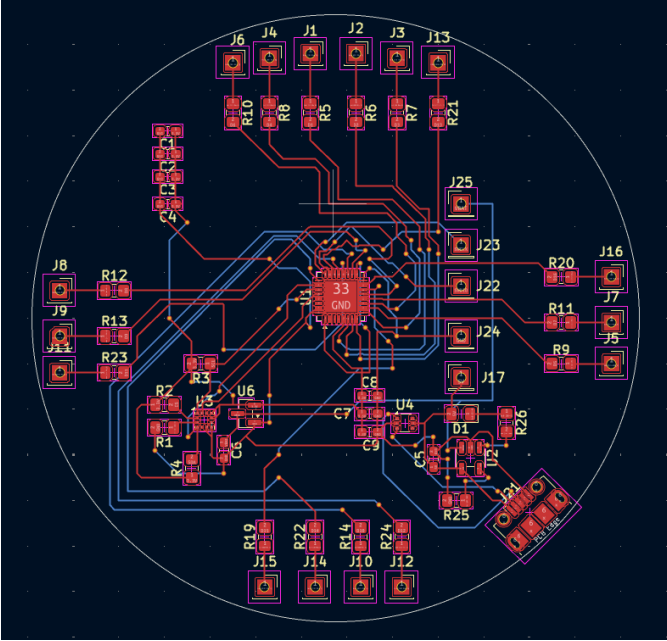


Fig. 3. Top view of the main electronics PCB with MCU, sensor, power management and LED connectors.

VI. SUMMARY

In this work we designed and implemented a compact wearable pendant that combines real-time physics-based fluid simulation with a custom LED matrix display. The system integrates three core components: (i) a FLIP-based fluid simulation algorithm adapted for execution on a microcontroller, (ii) a high-density LED matrix driven using a Charlieplexing

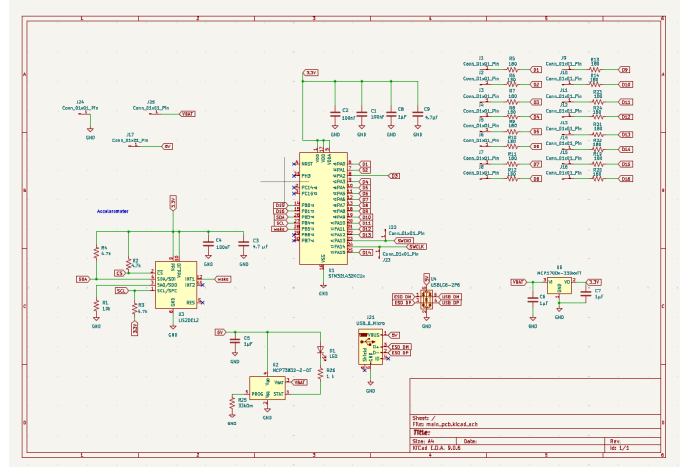


Fig. 4. Simplified schematic of the pendant electronics, including the LED driver network and power-management circuitry.

scheme to minimize the number of required I/O pins, and (iii) an inertial sensing unit that provides real-time motion data so that the simulated fluid responds naturally to gravity and user interaction.

During the hardware development stage, an initial approach based on a fully integrated multi-layer PCB was considered. However, practical limitations related to reliable microcontroller programming and debugging led to a revision of the hardware strategy. As a result, the final prototype adopts a modular architecture consisting of a standard STM32 development platform combined with a custom-designed LED matrix board.

From a system perspective, the separation of the LED matrix and the control electronics proved advantageous. It enabled independent optimization of the visual display and the control logic, while preserving the compact circular form factor required by the pendant design. The Charlieplexing technique allowed a large number of LEDs to be controlled with a limited number of microcontroller pins, making the approach suitable for wearable and space-constrained devices.

The mechanical enclosure was designed as a three-part 3D-printed case that aligns the LED matrix beneath a transparent dome, protects the electronics, and provides access for charging. The current prototype successfully demonstrates stable LED scanning, responsive motion sensing, and visually convincing fluid-like behavior.

Our work includes final integration of all components into the finished case, further optimization of power consumption, and refinement of the simulation parameters to improve visual smoothness and realism. And as a result, we have something similar to what we expected - pendant (but not compact) with real-time fluid simulation on LED-matrix.

VII. SOURCES

<https://mitxela.com/projects/fluid-pendant>
<https://www.youtube.com/watch?v=XmzBREkK8kY>
<https://www.youtube.com/watch?v=iKAVRgIrUOU>